

Post-Incident Review — DB Outage INC-20240915

Severity: P1 | Duration: 47 min | Impact: All production API endpoints

1. Summary

On 2024-09-15 at 14:23 UTC, the primary PostgreSQL cluster suffered a 47-minute complete outage. Root cause: a schema migration acquired an EXCLUSIVE lock on a 220M-row table, causing connection pool exhaustion and universal HTTP 503 responses.

2. Timeline

UTC Time	Event
14:18	Migration v2.14.1 initiated by automated deploy pipeline
14:19	EXCLUSIVE lock acquired on users table (220M rows)
14:23	Connection pool saturated; HTTP 503 on all endpoints
14:24	PagerDuty fires; on-call SRE paged
14:31	SRE identifies blocker via pg_stat_activity
14:33	Migration terminated; lock released
14:37	Connection pool recovers; APIs restored
14:45	Migration v2.14.1 rolled back
15:10	Post-incident review initiated

3. Root Cause

ALTER TABLE on a 220M-row table with NOT NULL DEFAULT rewrites the entire table under ACCESS EXCLUSIVE lock. No lock_timeout guard was in place.

```
ALTER TABLE users ADD COLUMN last_login_ip INET NOT NULL DEFAULT '0.0.0.0';
```

4. Remediation

Immediate: terminate migration, rollback, drain pool. Long-term: (1) mandatory zero-downtime review for tables >1M rows; (2) lock_timeout=5s on all prod migrations; (3) integrate pg_repack.